

LOT Pager Firmware

LOT® Pager — Firmware Specification (v0.1)

Target: ESP32-S3, ESP-IDF 5.x (C/C++), FreeRTOS. Repo path when created: `hardware/firmware/`.

1. Modules

Module	Role
net	Wi-Fi STA, TLS, device token storage (NVS encrypted)
lotapi	Calls the device API (see SOFTWARE-CONNECTOR)
ui	LVGL on GC9A01; screens: Notify, Idle(clock/weather), Status
alert	piezo/haptic patterns (soft, urgent)
button	debounce, short press = Copy, long press = dismiss/ack, 10 s = pairing mode
sense	BME688 (BSEC2), VEML7700, IMU; 5-min sampling
cam	on-demand capture only, JPEG ≤ 100 KB, upload on explicit action
power	Qi state, gauge, charger, sleep policy
ota	signed OTA via server, rollback on failed boot

2. States

PROVISION → PAIR → IDLE(light sleep) ⇌ NOTIFY ⇌ CAPTURE, with CHARGING overlay.

3. Timing

Poll 30 s on battery, 5 s on charger. Sensor post every 5 min, batched. Target ≥ 3 days on 120 mAh.

4. Notification handling

Payload: {id, text, level, expires_at}. Show text (≤ 60 chars), play pattern by level, wait for button. Short press → copy event with notification_id; long press → ack. Unacknowledged notifications re-alert once, then expire. Quiet hours enforced server-side and mirrored locally.

5. Security

Secure boot v2 + flash encryption; TLS pinned to server CA; token revocable from site; no inbound ports.

6. Camera / privacy

Camera never streams. Capture only on button chord or server request that the user approves on-device; screen shows a capture indicator.

7. Build / test

`idf.py build`; unit tests on host for parsing and state machine; HIL test via host CLI.